

Prompt Repetition in Low-Resource RTL Languages

Tests 1–2: baseline vs. repetition · replication of arXiv:2512.14982 · results as of 8 October 2026

Repetition significantly improved accuracy in 134 of 480 tests and hurt in 21 (McNemar, $p < 0.1$). Under the stricter exact test: 118 wins / 15 losses. On the original paper's tasks only: 71 / 17 of 336 (exact 65 / 11).

Setup

- **Methods:** baseline prompt vs. the same prompt repeated twice (paper formats A.3/A.4).
- **Models (8, non-reasoning):** Llama 3.2 1B / 3B, Llama 3.1 8B, Qwen 2.5 1.5B / 3B / 7B, Gemma-2 2B, Mistral 7B v0.3.
- **Languages (6, right-to-left):** Punjabi (Shahmukhi), Urdu, Pashto, Arabic, Persian, Sindhi.
- **Tasks (7):** ARC, OpenBookQA, CommonsenseQA, MGSM (GSM8K), NameIndex, MiddleMatch, ScriptMixed; MCQ tasks in question-first and options-first orderings.
- **Decoding:** temperature 0, max 100 output tokens, direct-answer system prompts. **280,816 generations, 0 errors.**
- **Statistics:** McNemar test without continuity correction, $p < 0.1$ (as in the paper), plus the exact binomial McNemar test as a robustness check.

Results by model (60 tests each)

Model	Wins	Losses	Exact W / L	Acc. baseline	Acc. repetition	Change
Llama 3.1 8B	26	1	24 / 1	37.4%	41.3%	+3.9 pt
Qwen 2.5 3B	23	1	19 / 0	26.4%	29.0%	+2.6 pt
Qwen 2.5 7B	21	0	20 / 0	32.8%	36.0%	+3.2 pt
Gemma-2 2B	14	2	11 / 2	25.5%	27.2%	+1.6 pt
Llama 3.2 3B	14	2	10 / 1	29.4%	31.1%	+1.7 pt
Llama 3.2 1B	14	7	14 / 4	19.6%	20.2%	+0.6 pt
Qwen 2.5 1.5B	11	2	10 / 2	22.1%	22.9%	+0.8 pt
Mistral 7B v0.3	11	6	10 / 5	21.5%	21.8%	+0.2 pt

By language

Language	Wins	Losses	Exact W / L	Tests
Arabic	38	4	35 / 2	80
Persian	34	2	28 / 2	80
Urdu	17	2	16 / 2	80
Pashto	17	3	15 / 2	80
Punjabi	15	6	14 / 3	80
Sindhi	13	4	10 / 4	80

By task

Task	Wins	Losses	Exact W / L	Tests
ScriptMixed	36	0	30 / 0	48
OpenBookQA*	31	2	31 / 1	96
CommonSenseQA	27	4	23 / 4	96
ARC*	20	5	19 / 3	96
GSM8K*	8	2	4 / 1	48
MiddleMatch*	7	7	7 / 5	48
NameIndex*	5	1	4 / 1	48

* task from the original paper. Accuracy is item-weighted over all languages, tasks and orderings.

Key findings

- Repetition helps consistently in low-resource RTL languages, most clearly on **multiple-choice** and **mixed-script (ScriptMixed)** inputs, where it never hurt.
- **Stronger models gain most:** Llama 3.1 8B and Qwen 2.5 3B / 7B improve by about 3–4 accuracy points with almost no losses; the weakest models (Llama 3.2 1B, Mistral 7B) barely move and lose more often.
- **Arabic and Persian** benefit most; Punjabi and Sindhi least.

- **MiddleMatch** (finding the name between two others in a list) is the only task where repetition does not help (wins equal losses).
- Effects are robust: most significant results survive the exact test.

Caveats

- **Machine translation:** Punjabi, Urdu and Arabic via Google Translate; Persian and Sindhi via NLLB-200 1.3B (two passes merged item by item with automatic QA; 12 defective items excluded). No native-speaker review yet.
- **Scoring:** retrieval answers use exact name matching after Perso-Arabic spelling normalisation (fuzzy matching was rejected because distinct names score above 0.80 similarity).
- **Model quirks:** Gemma-2's system prompt is merged into the user turn; Llama 3.x chat templates insert the current date.
- Model set and two extra tasks (CommonSenseQA, ScriptMixed) differ from the original paper.

Next steps

- Tests 3–4 (cross-lingual prompting), reusing all baseline / repetition results.
- Native-speaker review of prompts and translated benchmarks.

Data: raw generations on Kaggle (*fiznasir/pr-results-checkpoint*) and a private Hugging Face dataset; tables in *results/tests_1_2/* of the GitHub repository *FizzahNasir/prompt_repetition_research*.